

Tarek Halloul

Gameplay Programmer

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Professional Summary

Gameplay programmer with 4+ years of **Unity**-focused experience, including 2.5 years teaching game development and shipped projects across card, puzzle, arcade, educational, and **AR** genres. Skilled in **C#, C++, Unreal Engine 5**, and **Godot**, with hands-on work in character animation blending (*PyTorch/ONNX*), custom shaders, physics, AI, and AR integration (*Vuforia*). Comfortable across the full game development cycle, from prototyping and game feel to optimization and playable builds. Maintains 25+ public game development repositories on GitHub.

Core Skills

- **Game Engines:** Unity 2022 LTS, Unreal Engine 5.3 (Blueprints), Godot 4
- **Languages:** C#, C++, GDScript, Python
- **Gameplay Systems:** Physics, AI, animation systems & blending, shaders (ShaderLab), UI, input, game feel, object pooling, Scriptable Objects, Tilemap
- **AR & Tools:** Vuforia, Blender, Figma, Git, Agile/Scrum, profiling & optimization
- **ML Integration:** PyTorch, ONNX Runtime (animation / skeleton-based systems)

Professional Experience

Game Development Instructor — *GOMYCODE*

Aug 2021 – Jan 2024

- Delivered **Unity** game development curricula covering C#, physics, UI, and gameplay mechanics to young developers.
- Mentored students from prototype to playable builds, strengthening code-review and technical mentoring skills.

R&D Gameplay Programmer Intern — *Lanterns Studios*

Feb 2023 – Sep 2023

- Contributed to gameplay programming research within a commercial game development studio using **Unreal Engine**.
- Prototyped the **LMM** system to record multiple humanoid animations and blend them into a single animation driven by player decisions.

AR Developer Intern — *ENVAST*

Feb 2019 – Jun 2019

- Built an augmented reality reading app for the Kounouz publishing house using **Unity** and **Vuforia**.
- Designed AR interactions for a mobile app dedicated to children, from concept to working prototype.

Featured Projects

- **LMM** — Records multiple humanoid animations and blends them into a single animation that changes according to player decisions. *Unreal Engine 5, Unity, C#, C++, Python, PyTorch, ONNX Runtime*.
- **Chkobba (chkoba)** — 2D implementation of the Tunisian card game, building out the full gameplay loop and rules in Godot 4. *Godot 4, GDScript*.
- **My3DChess** — 3D chess game featuring custom shaders. *Unity, C#, ShaderLab*.
- **AR Reading App (Kounouz)** — Augmented reality experience for a children's reading app. *Unity, Vuforia, Firebase, Figma*.
- **DM Nova Prototype** — Interactive educational mobile app prototype for children. *Unity, Vuforia, Firebase, Figma*.
- **Arcade & Puzzle Suite** — 10+ playable titles: FlappyBird, Tetris, BrickBreaker, Arkanoid, Pong 2.0, Snake, Project 2048, Pacman, top-down shooter. *Unity, C#*.
- **Workshop & Game-Jam Demos** — Match-pairs flip game and Three-Cups logic game. *Unity, C#*.
- **FindLog (Divalto)** — Game created for Divalto during the Nuit de l'Info 2022 coding event. *Unity, C#*.
- **Android + Unity AR Integration** — Android application embedding Unity-based augmented reality features. *Unity, C#, Java*.

Education & Languages

B.S. Software Engineering — *EPI* (honors)

Sep 2020 – Sep 2023

Bachelor's in Computer Science — *ISITCOM* (honors)

Jan 2016 – Jan 2019

- **Languages:** Arabic (Native), English (Fluent), French (Fluent)